

Crop Yield Prediction Based on Weather and Soil Data

Data Cleaning, Data Handling & XGBoost Training Approach

Supporting document for Chapter Three (Methodology) — Maize (Zea mays L.)

Stage 1: Data Preparation

1. Overview

This document reports the data cleaning and data handling carried out for the maize yield prediction model, and explains how the data will be trained using XGBoost. It accompanies the Jupyter notebook *maize_yield_data_preparation.ipynb*, which contains the full, runnable code and output for every step described here.

The raw dataset (*crop_yield_dataset.csv*) contains 10,000 records and 13 columns covering four crops (maize, rice, barley and wheat). In line with the scope of the study, the data was restricted to maize, giving 2,477 records that form the basis of all subsequent work.

2. The Maize Dataset

After scoping to maize, each record describes one maize plot through the following variables, grouped by type:

Weather variables

- **Rainfall_mm** — total rainfall received (millimetres)
- **Temperature_C** — mean temperature (degrees Celsius)
- **Humidity_pct** — mean relative humidity (percent)

Soil variables

- **Soil_pH** — soil acidity/alkalinity (0–14 scale)
- **Soil_Type** — soil classification (Sandy, Loam, Clay)

Management variables

- **Fertilizer_Used_kg** — fertiliser applied (kilograms)
- **Pesticides_Used_kg** — pesticide applied (kilograms)
- **Planting_Density** — planting density of the plot
- **Irrigation** — irrigation method (Drip, Sprinkler, Flood, or Rainfed)

- **Previous_Crop** — crop grown the previous season (or none / fallow)
- **Region** — growing region (Region_A to Region_D)

Target variable: *Yield_ton_per_ha* — the observed maize yield in tonnes per hectare, which the model learns to predict.

3. Data Cleaning

Data cleaning removes structural and quality problems before the data is used. Three checks were applied.

3.1 Duplicate records

Exact duplicate rows can arise when the same plot is captured more than once during data extraction. All rows were compared and duplicates removed. Result: **0** duplicate maize records were found — the dataset contains no repeated rows.

3.2 Range and validity checks

Every numeric value was checked against a physically and agronomically possible range — for example, soil pH must fall between 0 and 14, humidity between 0 and 100 percent, and rainfall, fertiliser, pesticide, planting density and yield can never be negative. Any value failing these rules is treated as invalid and set to missing so it is handled uniformly in the next stage. Result: **0** out-of-range values were found — every maize value lies within its valid range.

3.3 Formatting consistency

Categorical labels (region, soil type, irrigation and previous crop) were stripped of stray spaces and standardised so that records group together correctly during encoding. Column data types were confirmed — numeric fields as numbers, category fields as text.

4. Data Handling: Missing Values

The most important handling decision in this dataset concerns the word **"None"**.

The Python *pandas* library, used to load the data, treats the text "None" as a missing value by default. In this dataset, however, "None" is a genuine category with a real meaning:

- **Irrigation = "None"** means the plot is rainfed — it has no irrigation system, which is real information, not a gap.
- **Previous_Crop = "None"** means no crop was grown the previous season (fallow / first cultivation) — again, a real fact about the plot.

If the default behaviour is accepted, the software wrongly reports 2,538 missing irrigation values and 2,031 missing previous-crop values across the full dataset. **Filling these "gaps" would overwrite genuine information and distort the data.**

The correct handling is to tell the loader that "None" is a real category, and only treat truly blank fields as missing. The "None" labels were then renamed to *Rainfed* and *No_Previous_Crop* so their meaning is unmistakable. After this correction, the number of genuinely missing cells in the dataset is **0**.

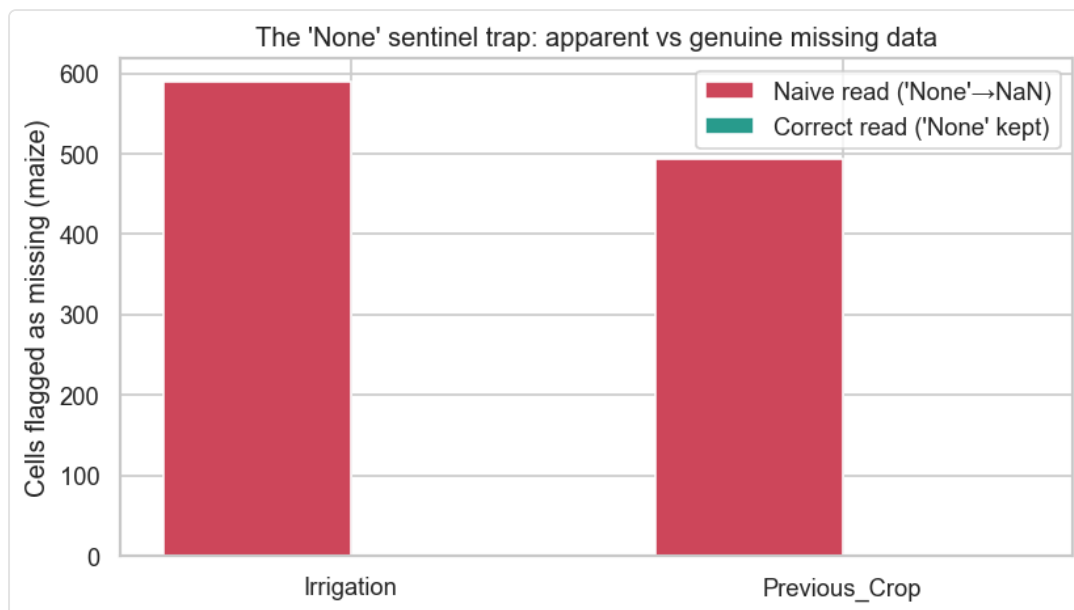


Figure 1. Apparent missing data under a naive read versus the true picture once "None" is correctly kept as a category.

4.1 Missing-data policy

Although this dataset has no genuine gaps after the correction above, a robust handling routine was still put in place so the pipeline stays reliable if new or updated data is added later:

- Any record missing more than half of its fields is removed, as it carries too little reliable information.
- Remaining gaps in numeric variables are filled with the column median, which resists extreme values better than the mean.
- Remaining gaps in categorical variables are filled with the most frequent category (the mode).

5. Data Preprocessing

5.1 Encoding categorical variables

Machine-learning models require numeric input, so the four categorical columns were converted using one-hot encoding, which creates a separate 0/1 indicator for each category. After encoding, the maize dataset has **19** input features.

5.2 Normalisation (standardisation)

Because the variables are measured on very different scales (for example rainfall in the hundreds of millimetres versus soil pH between 0 and 14), each numeric feature was standardised to a mean of 0 and a standard deviation of 1 using the formula $z = (x - \mu) / \sigma$. The mean (μ) and standard deviation

(σ) are calculated from the training data only and then applied to the test data. This prevents information from the test set leaking into training, which would otherwise make the model look more accurate than it really is.

5.3 Training and test split

The prepared data was divided into an 80% training set (1,981 records) and a 20% test set (496 records). The split is stratified by region, so every maize-growing region is represented in the same proportion in both sets. The model learns only from the training set; the test set is held back for a fair, final evaluation.

6. What the Data Shows

Exploratory analysis of the maize data highlights which factors move yield. Fertiliser applied is by far the strongest single driver, followed by rainfall; the other variables have weaker individual relationships with yield. This is agronomically sensible and gives the model a clear signal to learn from.

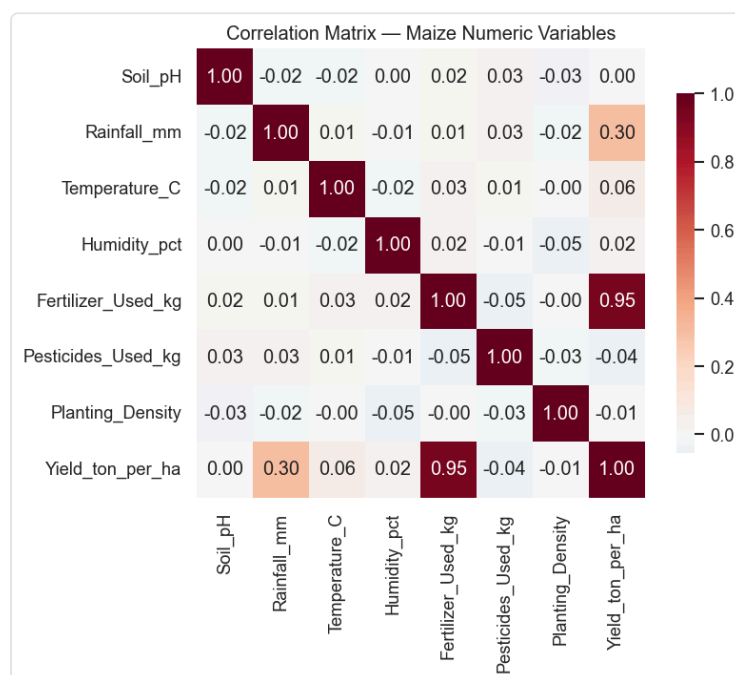


Figure 2. Correlation between maize numeric variables and yield. Fertiliser (0.95) and rainfall (0.30) are the dominant drivers.

7. How the Data Will Be Trained with XGBoost

XGBoost (Extreme Gradient Boosting) is one of the two main models in this study, alongside Random Forest, with Linear Regression as the baseline.

7.1 How XGBoost works (in brief)

XGBoost builds a sequence of small decision trees. The first tree makes a rough prediction of yield; each following tree is trained specifically to correct the errors left by the trees before it. Adding many

such trees together, step by step, produces a highly accurate combined prediction. Two features make XGBoost especially well suited to this kind of tabular farm data:

- **Error-correcting (boosting):** each new tree focuses on the records the model is currently getting wrong, so accuracy improves with every step.
- **Built-in regularisation:** XGBoost adds a penalty for overly complex trees, which stops it from simply memorising the training data and helps it generalise to new plots.

7.2 Training procedure

- Input: the 19 preprocessed maize features (weather, soil and management). Target: yield in tonnes per hectare.
- The model is trained on the 80% training set only, using a fixed random seed so results are reproducible.
- Key settings control learning: the number of trees, the learning rate (how much each tree contributes), the maximum tree depth, and sampling of rows and columns per tree to reduce overfitting. These are tuned in the next stage using cross-validation.
- The same training set and features are used for Linear Regression and Random Forest, so the three models can be compared fairly.

7.3 Evaluation

Each trained model is scored on the held-back 20% test set using four metrics: Root Mean Square Error (RMSE) and Mean Absolute Error (MAE), both in tonnes per hectare and smaller-is-better; R-squared, the share of yield variation explained (closer to 1 is better); and accuracy, the percentage of predictions within an acceptable margin of the true yield.

7.4 Preliminary result

A first, untuned XGBoost model was run to confirm the prepared data trains successfully. On the maize test set it achieved:

Metric	Preliminary XGBoost (untuned)
RMSE	5.843 ton/ha
MAE	4.641 ton/ha
R-squared	0.9766
Accuracy (within ± 11.57 ton/ha)	94.56%

These are strong early numbers (R-squared ≈ 0.98) and confirm the data is model-ready. *They are preliminary — full hyper-parameter tuning and the complete comparison of Linear Regression, Random Forest and XGBoost belong to the next project stage.*

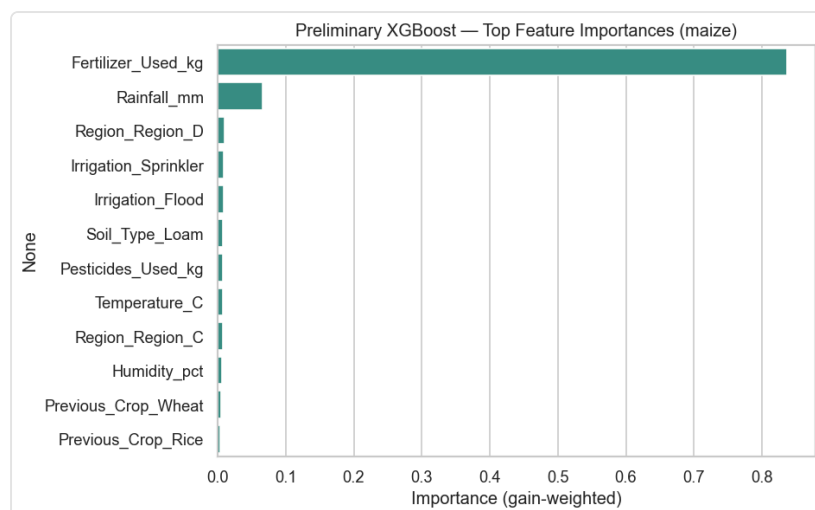


Figure 3. Variables the preliminary XGBoost model relied on most — led by fertiliser and rainfall.

8. Note on Alignment with Chapter Three

One point to reconcile in the written chapter. Chapter Three currently describes a dataset integrated from ERA5 (weather), SoilGrids 2.0 (soil) and FAOSTAT (yield), listing variables such as solar radiation, wind speed, nitrogen, phosphorus, potassium, organic carbon and soil moisture. The dataset actually supplied (*crop_yield_dataset.csv*) is a single, already-combined file whose maize variables are those listed in Section 2 above. It does not contain those specific weather/soil measurements, and it adds useful management variables (fertiliser, pesticides, irrigation, planting density, previous crop).

This does not weaken the methodology — every stage (cleaning, missing-data handling, encoding, standardisation, the 80/20 stratified split, and the Linear Regression / Random Forest / XGBoost comparison) applies exactly as written. It is recommended that the variable lists and data-source wording in Chapter Three be adjusted to match the actual dataset, or that a dataset matching the current wording be sourced if those specific variables are required. Either way, the pipeline in this document and the accompanying notebook is complete and reproducible.

9. Deliverables for This Stage

- *maize_yield_data_preparation.ipynb* — fully documented, runnable notebook of the whole pipeline.
- *outputs/maize_clean.csv* — the cleaned maize dataset (human-readable).
- *outputs/maize_model_ready.csv* — the encoded, standardised, split-tagged dataset ready for modelling.
- *outputs/figures/* — all charts used in this report.
- *outputs/metrics.json* — every number quoted here, for exact reproducibility.

